A PROJECT REPORT ON

Financial Performance Analysis of State Bank of India (FY2017-FY2026) Using Python

Supervised By

Dr. Anirban Nandy

ASSISTANT PROFESSOR

Centre for Innovative Studies (CIS)

University of North Bengal

Co-guide

Subhajit Saha

FACULTY

Centre for Innovative Studies (CIS)

University of North Bengal

Submitted in partial fulfilment of the requirements for the award of the

DIPLOMA IN DATA SCIENCE (DDS) COURSE 2025-26

UNDER THE CENTRE FOR INNOVATIVE STUDIES (CIS)

UNIVERSITY OF NORTH BENGAL

Submitted By

SAGNIK DUTTAGUPTA

ROLL NO: 22325009

REGISTRATION NUMBER: 0402412020694

SESSION 2025-2026

CENTRE FOR INNOVATIVE STUDIES (CIS)

UNIVERSITY OF NORTH BENGAL RAJA RAMMOHUNPUR, DARJEELING, 734013 WEST BENGAL, INDIA 2026

CENTRE FOR INNOVATIVE STUDIES
UNIVERSITY OF NORTH BENGAL

Raja Rammohunpur, District: Darjeeling, West Bengal, Pin-734013, India

CERTIFICATE

This is to certify that the project report entitled "Financial Performance Analysis of State Bank of India (FY2017-FY2026) Using Python" is a bonafide record of the work carried out by Sagnik Duttagupta. (Roll No. 22325009, Registration No. 0402412020694) in partial fulfilment of the requirements for the Diploma in Data Science (Project Work II, DDS-293) at the Centre for Innovative Studies, University of North Bengal, during the session [2025-26].

The work was carried out under my supervision and guidance. To the best of my knowledge, it is original and has not been submitted to any other institution or university for the award of any degree or diploma.

I wish the student all success in future endeavours.

Date: Supervisor

Place: University of North Bengal

Dr. Anirban Nandy
Assistant Professor
Centre for Innovative Studies
University of North Bengal

Co-guide

Subhajit Saha
Faculty
Centre for Innovative Studies
University of North Bengal

DECLARATION

I, Sagnik Duttagupta, bearing Registration No. 0402412020694, a student of Diploma in Data Science (DDS) at Centre for Innovative Studies, University of North Bengal, hereby declare that the project report entitled "Financial Performance Analysis of State Bank of India (FY2017-FY2026) Using Python" is a record of my own work, carried out under the guidance of Dr. Anirban Nandy, Assistant Professor, Centre for Innovative studies, University of North Bengal and Co-guidance of Sir Subhajit Saha, Faculty, Centre for Innovative studies, University of North Bengal.

Wherever I used material from other sources it has been acknowledged in the reference section. This report has not been submitted earlier, in whole or in part, to any other institute or university for the award of any diploma or degree.

Signature
Name: Sagnik Duttagupta
Date:
Place: University of North Bengal

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who helped me to complete this project successfully.

First of all, I thank my project guide, Dr. Anirban Nandy, for the valuable guidance, constant encouragement and useful suggestions given throughout the process of the project work. The regular feedback helped me to improve this report. I am also thankful to Sir Subhajit Saha for his valuable teachings, guidance and directions provided.

I am also thankful to the Head of the Department and all the faculty members of Centre for Innovative Studies, University of North Bengal for teaching me the fundamentals of data science and programming that I have applied in this project. I thank the institute for providing the computer lab and the other facilities I needed.

I am grateful to the creators of the open-source tools used in this project, especially Google Colab, and to the authors of the free online documentation that I used while learning and testing. Finally,

I thank my family and friends for their support and motivation.

Sagnik Duttagupta

ABSTRACT

What the project is: The project is a systematic review of the financial performance of the State Bank of India (SBI) over a ten-year period from FY2017 to FY2026.

Why it was developed: The study was developed to evaluate how SBI achieved sustainable profitability without excessive financial leverage by assessing key areas including top-line scale, cost structure, balance sheet composition, cash flows, and shareholder value. It aims to determine the structural drivers of SBI's turnaround from a period of severe financial distress and net losses into an era of record profitability and asset utilization.

The technologies used: The statistical programming language Python was utilized to conduct the financial analysis and minimize statistical base-effect distortions. Python-based techniques included applying Ordinary Least Squares (OLS) regression to map the fundamental relationship between Earnings Per Share (EPS) and market share price, as well as utilizing statistical dispersion tools like the Interquartile Range (IQR) and Coefficient of Variation (CV) to measure quarterly profit-margin stability.

The result: The result is a comprehensive quantitative analysis report demonstrating SBI's structural transition from a net loss in FY2018 to a record net profit of ₹83.30 thousand crores by FY2026, alongside a 123.5% growth in total revenue. The analysis revealed that the bank successfully reached the 1.0% Return on Assets (ROA) gold standard benchmark in FY2024 and executed an extraordinary de-risking phase that dropped its Borrowings-to-Net Worth ratio to a decade-low of 11.44 times by FY2026. Finally, the report shows that these operational efficiency and productivity gains drove EPS to ₹90.2 and sparked a 234.1% appreciation in the market share price over the ten-year period.

Keywords: Financial Analysis, Python, State Bank of India (SBI), Profitability Metrics, Return on Assets (ROA), Ordinary Least Squares (OLS) Regression, Balance Sheet Management

TABLE OF CONTENTS

CERTIFICATE……..……...…. **i**
DECLARATION……..…… **ii**
ACKNOWLEDGEMENT……...……...**iii**
ABSTRACT……...…. **iv.**

1. INTRODUCTION

1.1 Background and Context **1**
1.2 Motivation of the Study **1**
1.3 Objectives **1**
1.4 Scope and Methodology **2**

2. LITERATURE REVIEW

2.1 Theoretical Foundations and Overview of State Bank of India (SBI) **3**
2.2 Previous Studies on SBI's Financial Performance, Growth and Profitability **3**
2.3 Asset Quality, Non-Performing Assets (NPAs), Cost Efficiency and Balance Sheet Management **3–4**
2.4 Application of Python in Financial Analysis, Summary of Literature and Research Gap **4**

3. GROWTH AND PROFITABILITY

3.1 Year-on-Year Growth in Sales and Net Profit (FY2017–FY2026) **5–6**
3.2 CAGR of Sales, Net Profit, and Total Assets **6–7**
3.3 Trend in Net Profit Margin **7–8**
3.4 Changes in Return on Equity (ROE) and Return on Assets (ROA) **8–9**

4. INCOME AND COST STRUCTURE

4.1 Annual Share of Interest, Employee Costs, and Other Expenses in Sales **10–11**
4.2 Contribution of Other Income to Sales **11–12**

5. BALANCE SHEET AND CASH FLOW

5.1 Changes in Asset and Liability Composition **13–15**
5.2 Operating, Investing, and Financing Cash Flow Analysis **15–16**

6. SHAREHOLDER RETURNS

6.1 Trends in Earnings Per Share (EPS), Price-to-Earnings (P/E), and Price-to-Book (P/B) Ratios **17–19**
6.2 Trends in Dividends and Dividend Payout Ratio **19–20**
6.3 Relationship Between Share Price and EPS **20–21**

7. QUARTERLY ANALYSIS

7.1 Trends in Quarterly Sales, Operating Profit, and Net Profit **22–23**
7.2 Quarterly Profit Margin Stability **24**

8. CONCLUSION, LIMITATIONS AND FUTURE SCOPE

8.1 Conclusion **25–26**
8.2 Limitations **26–27**
8.3 Future Scope **27–28**

1. INTRODUCTION

1.1 Background and Context

The Indian banking sector has navigated a complex macroeconomic landscape over the past decade, characterized by severe asset quality cycles, systemic consolidation, and regulatory tightening, eventually giving way to a robust credit super-cycle. As India's premier public sector commercial bank, the State Bank of India (SBI) serves as a critical barometer for the broader financial system. During the late 2010s, the institution absorbed massive structural shocks, including the integration of its associate banks and a rigorous corporate Non-Performing Asset (NPA) cleanup. Emerging from this period of distress, SBI has leveraged modern digital banking architecture, disciplined liability pricing, and rigorous underwriting standards to transition into an era of secular compounding, marked by fortified capital buffers and sustained high-margin retention.

1.2 Motivation of the Study

While traditional accounting frameworks provide static snapshots of financial health, understanding the true anatomy of a banking turnaround requires a dynamic, time-series evaluation. The transition of SBI from reporting a net loss in FY2018 to achieving record profitability and crossing the 1.0% Return on Assets (ROA) benchmark by FY2024 represents a remarkable case study in balance sheet management and operating leverage. Analyzing this evolution through a data-driven lens reveals how structural shifts in funding costs, employee productivity, and asset deployment interact across different monetary policy regimes to generate shareholder wealth.

1.3 Objectives

The primary objectives of this systematic review are:

- To evaluate the year-on-year growth trajectory and compound annual growth rates (CAGR) of core sales, net profit, and total assets from FY2017 to FY2026.
- To dissect the bank's income and cost architecture, quantifying the impact of operating leverage and the counter-cyclical buffering role of non-interest income.
- To assess the structural fortification of the balance sheet, explicitly tracking the de-escalation of financial leverage and shifts in cash flow regimes.
- To establish the empirical relationship between fundamental earnings growth (EPS) and market valuation multiples, alongside evaluating the bank's dividend distribution policy.
- To measure the stability and statistical dispersion of profit margins across recent quarterly reporting cycles.

1.4 Scope and Methodology

This study is confined to the standalone financial performance of the State Bank of India over a ten-year horizon (FY2017 to FY2026), incorporating both annual historical data and a recent ten-quarter high-frequency analysis (March 2024 to June 2026). Bridging commercial accounting acumen with programmatic data science workflows, the analysis heavily utilizes Python-based exploratory data analysis (EDA) executed within Jupyter and Google Colab environments. By deploying libraries such as Pandas and Matplotlib for data aggregation and visualization, the study systematically models multi-year financial metrics, calculates variance and coefficients of dispersion, and applies ordinary least squares (OLS) regression to quantify valuation dynamics.

2. LITERATURE REVIEW

2.1. Theoretical Foundations and Overview of State Bank of India (SBI)

- **Theoretical Foundations:** The foundational role of commercial banks is to mobilize savings and provide credit to support economic growth. Theoretical evaluation of a bank's financial performance relies on top-line scale, profitability metrics (such as Net Profit Margin, Return on Equity, and Return on Assets), income structure, balance sheet leverage, cash flow, and shareholder valuation. A Return on Assets (ROA) of 1% is widely considered the theoretical gold standard and benchmark for banking institutions.
- **Overview of SBI:** State Bank of India underwent severe financial stress in FY2018, reporting a net loss of ₹4.56 thousand crores following the merger of five associate banks and Bharatiya Mahila Bank. Following this consolidation, the bank executed a structural turnaround driven by its CASA deposit base, expansive branch network, and digital platforms like YONO. By FY2026, SBI's interest revenue expanded to ₹514.90 thousand crores, and its net profit reached a record ₹83.30 thousand crores.

2.2. Previous Studies on SBI's Financial Performance, Growth and Profitability

- **Performance Metrics:** The systematic review of SBI's performance emphasizes analyzing the year-on-year growth and Compound Annual Growth Rates (CAGR) of sales, net profits, and total assets. Performance evaluations highlight a massive divergence between SBI's 9-year baseline net profit CAGR of 91.4% (distorted by a depressed FY2017 base) and its 5-year normalized CAGR of 30.0%.
- **Profitability Transitions:** Financial assessments segment the bank's recent history into distinct eras: a sub-cost-of-capital era (FY2017–FY2021) where ROE remained below the 10-year mean, and a value creation era (FY2022–FY2025) where ROE crossed double digits and ROA successfully reached the 1.0% benchmark.
- **Margin Evolution:** Studies on margin stability show SBI transitioning from negative profit margins (-2.0% in FY2018) to a sustained high-margin plateau, stabilizing at a Net Profit Margin of 16.2% by FY2026.

2.3. Asset Quality, Non-Performing Assets (NPAs), Cost Efficiency and Balance Sheet Management

- **Asset Quality & NPAs:** The Indian banking sector between FY2017 and FY2019 was heavily characterized by public sector banks facing high Non-Performing Assets (NPAs) and increased provisioning. For SBI, heavy provisioning for legacy corporate NPAs between FY2018 and FY2020 severely constrained net profit retention and eroded net worth growth.

- **Cost Efficiency:** Analyses of cost structures show that SBI's interest expense to sales dropped during pandemic-era monetary easing before rising to 61.2% in FY2026 due to deposit competition. Employee costs peaked at 21.2% in FY2022 due to merger integrations and wage revisions, but subsequent productivity and digital efficiency gains reduced this burden to 14.1% by FY2026.
- **Balance Sheet Management:** Effective balance sheet management is tracked through financial leverage multiples. SBI successfully executed an extraordinary de-risking phase post-FY2021, dropping its Borrowings-to-Net Worth multiple from a pandemic peak of 15.06x to a decade-low of 11.44x in FY2026.

2.4. Application of Python in Financial Analysis, Summary of Literature and Research Gap

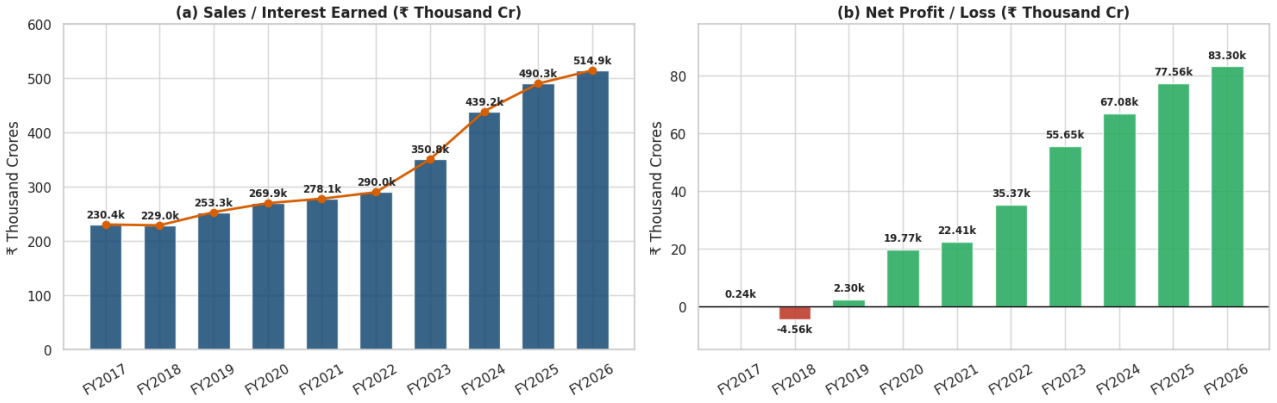
- **Application of Python:** Integrating computational data science into financial reviews allows analysts to identify long-term structural trends and minimize distortions caused by unusual earnings periods. Python-based techniques utilized include isolating statistical base-effect distortions by comparing multi-year CAGRs, and utilizing Ordinary Least Squares (OLS) regression to map the fundamental relationship between Earnings Per Share (EPS) and market share price.
- **Statistical Dispersion:** Python is also applied to measure quarterly profit-margin stability using statistical dispersion tools like the Interquartile Range (IQR) and Coefficient of Variation (CV).
- **Research Gap:** The primary research gap identified is understanding the exact structural drivers of SBI's turnaround: specifically, whether the bank's post-FY2021 profit growth and 234.1% stock-price rally were driven by genuine operational efficiency and asset utilization, or if they were merely the result of valuation expansion and financial leverage. Furthermore, limitations in standard aggregated financial data point to a need for future studies involving more granular banking-spread analyses and multivariate econometric modeling.

3. GROWTH AND PROFITABILITY

3.1. Year-on-Year Growth in Sales and Net Profit (FY2017–FY2026)

Over the ten-year horizon spanning FY2017 to FY2026, State Bank of India (SBI) underwent a structural transformation from a period of balance-sheet stress and consolidation to sustained top-line expansion and record profitability.

Financial Year	Sales / Interest Earned (₹ Thousand Cr)	YoY Sales Growth (%)	Net Profit / Loss (₹ Thousand Cr)	Post-Turnaround Net Profit YoY Growth (%)
FY2017	230.40	—	0.24	—
FY2018	229.00	-0.6%	-4.56	—
FY2019	253.30	10.6%	2.30	—
FY2020	269.90	6.5%	19.77	—
FY2021	278.10	3.1%	22.41	13.3%
FY2022	290.00	4.3%	35.37	57.9%
FY2023	350.80	21.0%	55.65	57.3%
FY2024	439.20	25.2%	67.08	20.6%
FY2025	490.30	11.6%	77.56	15.6%
FY2026	514.90	5.0%	83.30	7.4%



1. Sales / Interest Earned Trajectory

- **Consolidation & Pandemic (FY2017–FY2022):** Moved from ₹230.4k cr (FY2017) through a cleanup dip in FY2018 (₹229.0k cr), recovering moderately to ₹290.0k cr by FY2022 (mean YoY growth: 9.63%).
- **High-Growth Expansion (FY2023–FY2025):** Surged due to strong credit off-take and higher interest rates, peaking with 25.2% YoY growth in FY2024 (₹439.2k cr) and reaching ₹490.3k cr in FY2025.
- **Maturation (FY2026):** Growth normalized to 5.0%, crossing the half-trillion mark at **₹514.9 thousand crores**.

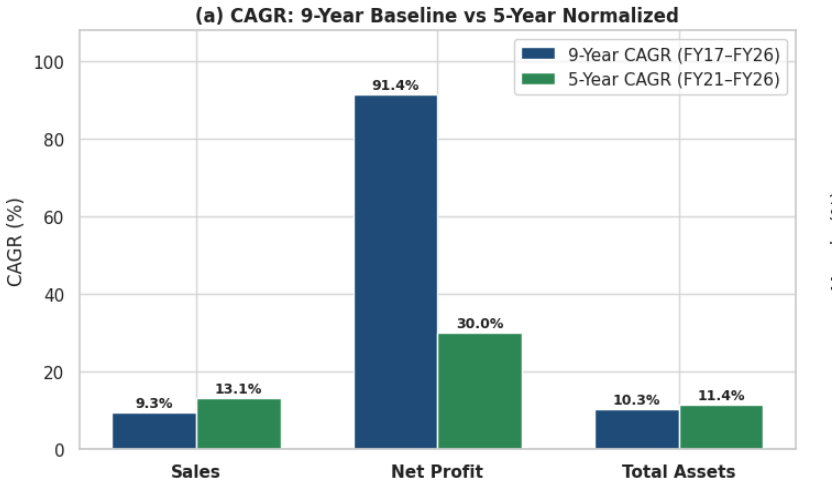
2. Net Profit Trajectory & Turnaround

- **Trough & Recovery (FY2017–FY2020):** Hit a low with a net loss of ₹-4.56k cr in FY2018 due to NPA provisioning, before a major turnaround and breakout to ₹19.77k cr by FY2020.

- **Post-Turnaround Compounding (FY2021–FY2023):** Accelerated rapidly with massive YoY surges—57.9% in FY2022 and 57.3% in FY2023—driven by falling credit costs and expanding margins.
- **High-Base Stabilization (FY2024–FY2026):** Continued double-digit growth before normalizing to 7.4% in FY2026, reaching an all-time high of **₹83.30 thousand crores**.

3.2. CAGR of Sales, Net Profit, and Total Assets

Evaluating SBI's Compound Annual Growth Rate (CAGR) across two distinct timeframes—a 9-year baseline (FY2017–FY2026) and a 5-year normalized post-turnaround window (FY2021–FY2026)—isolates the statistical base-effect distortions of the FY2017–FY2018 asset quality cycle.



Financial Metric	9-Year Baseline CAGR (FY2017–FY2026)	5-Year Normalized CAGR (FY2021–FY2026)	Structural Interpretation
Sales (Interest Earned)	9.3%	13.1%	Accelerated core revenue generation post-FY2021
Net Profit	91.4%	30.0%	Transition from low-base recovery to strong organic compounding
Total Assets	10.3%	11.4%	Consistent, disciplined balance-sheet expansion

Sales CAGR (9.3% vs. 13.1%)

- **Growth Acceleration:** The 5-year normalized sales CAGR (13.1%) exceeds the 9-year baseline (9.3%) by 380 basis points.
- **Driver:** Reflects sluggish top-line growth in the early years (FY2017–FY2021) versus vigorous loan book expansion and higher asset yields in the later period (FY2021–FY2026).

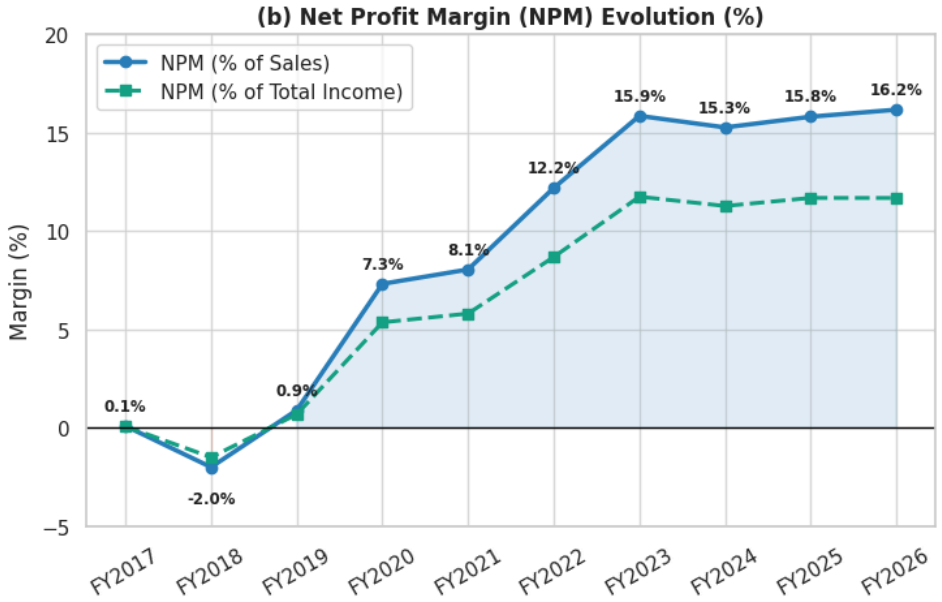
Net Profit CAGR (91.4% vs. 30.0%)

- **Distorted Baseline:** The 9-year Net Profit CAGR is an extraordinary 91.4% due to a depressed base year in FY2017 (₹0.24 thousand crores).
- **Normalized Growth:** The 5-year normalized CAGR sits at a robust 30.0% (using a FY2021 base of ₹22.41 thousand crores).
- **Leverage:** In both windows, Net Profit CAGR significantly outstripped Sales CAGR, highlighting powerful operating and credit-cost leverage.

Total Assets CAGR (10.3% vs. 11.4%)

- **Steady Expansion:** Balance sheet growth remained consistent, compounding at 10.3% over 9 years and accelerating slightly to 11.4% over the final 5 years.
- **Asset Efficiency:** In the 5-year normalized window, Sales CAGR (13.1%) outpaced Total Assets CAGR (11.4%), pointing to improved asset utilization and higher earning-asset yields.

3.3. Trend in Net Profit Margin

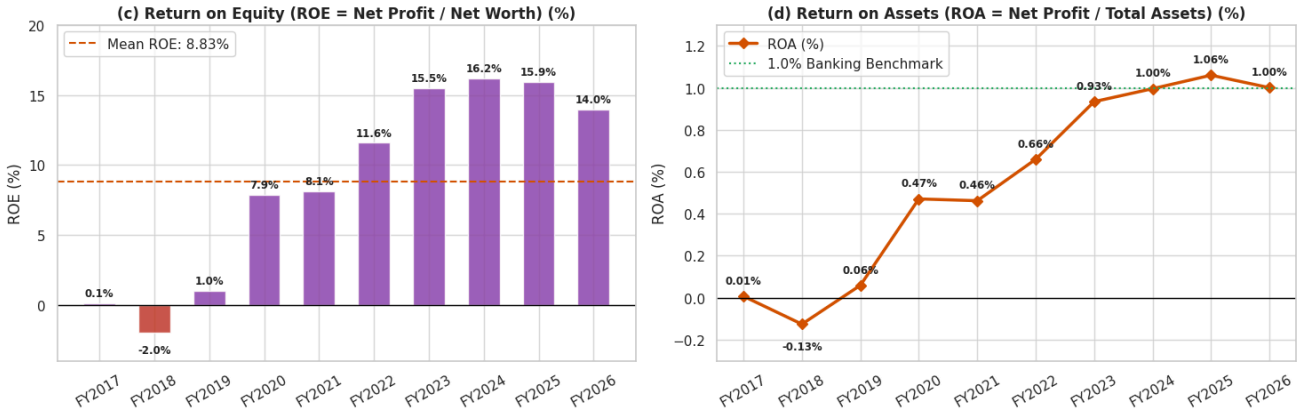


The evolution of SBI's Net Profit Margin (NPM)—tracked both as a percentage of Sales (Interest Earned) and as a percentage of Total Income (Interest Earned plus Non-Interest Income)—illustrates the bank's shift from value erosion to sustained high-margin retention.

- **Distress and Inflection (FY2017–FY2019):** In FY2017, NPM (% of Sales) stood at a negligible 0.1%, before dipping into negative territory at -2.0% in FY2018 due to heavy provisioning expenses. FY2019 marked the operational inflection point, with NPM recovering to 0.9%.
- **Structural Re-rating (FY2020–FY2023):** As legacy bad loans were fully provided for and credit costs normalized, a significantly larger portion of top-line revenue flowed directly to the bottom line. NPM (% of Sales) jumped to 7.3% in FY2020 and 8.1% in FY2021, before surging to 12.2% in FY2022 and 15.9% in FY2023.
- **High-Margin Plateau (FY2024–FY2026):** From FY2023 onward, SBI locked in its structural efficiency gains. NPM (% of Sales) stabilized within a narrow, elevated band: 15.9% in FY2023, 15.3% in FY2024, 15.8% in FY2025, and peaking at 16.2% in FY2026.

- **Sales vs. Total Income Margin Spread:** The trajectory of NPM (% of Total Income) closely mirrors NPM (% of Sales) at a proportionally lower level—rising from near zero in FY2017 and below zero in FY2018 to roughly 5.5%–5.8% in FY2020–FY2021, before stabilizing near the 11.3%–11.8% range between FY2023 and FY2026. The widening absolute spread between the two curves after FY2021 reflects the growing scale of core interest income alongside stable non-interest fee and treasury income streams.

3.4. Changes in Return on Equity (ROE) and Return on Assets (ROA)



Return on Equity ($ROE = \frac{Net\ Profit}{Net\ Worth}$) and Return on Assets ($ROA = \frac{Net\ Profit}{Total\ Assets}$) confirm that SBI's profit surge was driven by genuine asset productivity rather than excessive financial leverage.

Financial Year	Net Profit Margin (% of Sales)	Return on Equity (ROE %)	Return on Assets (ROA %)	Position vs. 1.0% ROA Benchmark
FY2017	0.1%	0.1%	0.01%	-99 bps below benchmark
FY2018	-2.0%	-2.0%	-0.13%	-113 bps below benchmark
FY2019	0.9%	1.0%	0.06%	-94 bps below benchmark
FY2020	7.3%	7.9%	0.47%	-53 bps below benchmark
FY2021	8.1%	8.1%	0.46%	-54 bps below benchmark
FY2022	12.2%	11.6%	0.66%	-34 bps below benchmark
FY2023	15.9%	15.5%	0.93%	-7 bps below benchmark
FY2024	15.3%	16.2%	1.00%	At 1.00% benchmark
FY2025	15.8%	15.9%	1.06%	+6 bps above benchmark
FY2026	16.2%	14.0%	1.00%	At 1.00% benchmark

Return on Equity (ROE) Progression

- **Sub-Cost-of-Capital Era (FY2017–FY2021):** ROE was minimal initially (0.1% in FY2017, -2.0% in FY2018) and remained below the 10-year mean of 8.83% despite a recovery to 7.9%–8.1% in FY2020–FY2021.

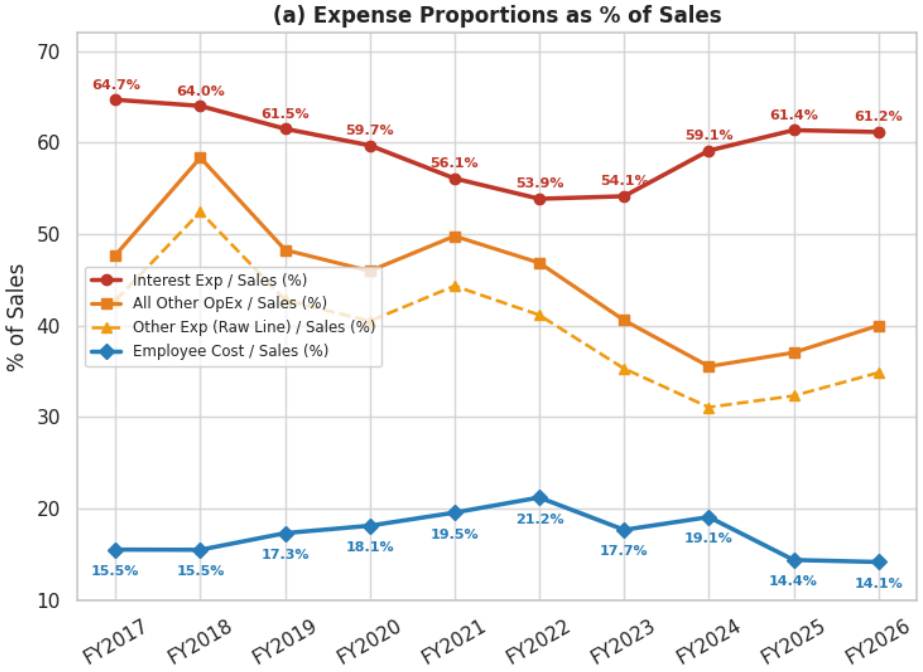
- **Value Creation Era (FY2022–FY2025):** Crossed double digits at 11.6% in FY2022, surged to a peak of 16.2% in FY2024, and held steady at 15.9% in FY2025.
- **Capital Base Expansion (FY2026):** ROE moderated to 14.0% as strong retained earnings expanded the net worth denominator faster than net profit growth, reinforcing the bank's core equity buffer.

Return on Assets (ROA) Progression

- **Climbing from Trough (FY2017–FY2022):** ROA reflected early distress (0.01% in FY2017, -0.13% in FY2018) before climbing to a plateau of 0.46%–0.47% and stepping up to 0.66% by FY2022.
- **Reaching the 1.0% Gold Standard (FY2023–FY2026):** Approached the benchmark at 0.93% in FY2023, hit 1.0% in FY2024, peaked at 1.06% in FY2025, and maintained 1.0% in FY2026.
- **Underwriting Discipline:** Sustaining a ≥1.0% ROA across three consecutive years while total assets grew by 11.4% highlights superior asset quality and cost efficiency

4. INCOME AND COST STRUCTURE

4.1. Annual Share of Interest, Employee Costs, and Other Expenses in Sales



An examination of State Bank of India's (SBI) cost architecture relative to its core revenue (Sales / Interest Earned) from FY2017 to FY2026 reveals a marked shift in liability pricing, employee productivity, and overhead absorption across distinct monetary policy and credit cycles.

Financial Year	Interest Exp / Sales (%)	Employee Cost / Sales (%)	All Other OpEx / Sales (%)	Other Exp (Raw Line) / Sales (%)	Implied Gross Interest Spread (NII / Sales %)
FY2017	64.7%	15.5%	~47.7%	~42.5%	35.3%
FY2018	64.0%	15.5%	~58.5%	~52.5%	36.0%
FY2019	61.5%	17.3%	~48.3%	~42.6%	38.5%
FY2020	59.7%	18.1%	~46.1%	~40.7%	40.3%
FY2021	56.1%	19.5%	~49.8%	~44.4%	43.9%
FY2022	53.9%	21.2%	~46.9%	~41.3%	46.1%
FY2023	54.1%	17.7%	~40.7%	~35.4%	45.9%
FY2024	59.1%	19.1%	~35.6%	~31.1%	40.9%
FY2025	61.4%	14.4%	~37.2%	~32.4%	38.6%
FY2026	61.2%	14.1%	~40.0%	~35.0%	38.8%

Interest Expended to Sales (Cost of Funds)

- **High-Cost Legacy Era (FY2017–FY2018):** High interest expenses (64.0%–64.7% of Sales) restricted the net interest spread to 35.3%–36.0%.
- **Monetary Easing & CASA Optimization (FY2019–FY2022):** Pandemic rate cuts and surplus liquidity reduced interest costs to 53.9% of Sales by FY2022, expanding the NII spread to 46.1%.
- **Policy Tightening & Repricing (FY2023–FY2026):** Deposit competition and rate hikes pushed the ratio back up to 61.2% by FY2026—still 3.5 percentage points lower than FY2017.

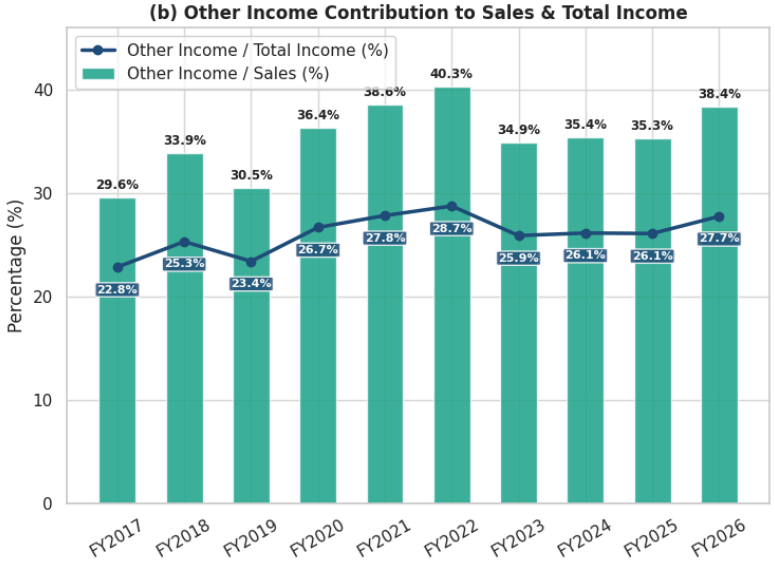
Employee Costs to Sales (Workforce Leverage)

- **Upward Creep (FY2017–FY2022):** Rose from 15.5% to 21.2% due to associate bank integration, wage revisions, and slower income growth during the pandemic.
- **Productivity Gains & Digital Efficiency (FY2023–FY2026):** Despite a temporary wage-revision spike in FY2024 (19.1%), employee costs dropped to **14.1% by FY2026** due to strong productivity and digital scaling.

All Other Operating and Non-Interest Expenses to Sales

- **Peak Burden (FY2017–FY2018):** Surged to 58.5% in FY2018 due to heavy provisioning, contributing to a net loss that year.
- **Normalization & Relief (FY2019–FY2024):** Declined steadily to 35.6% by FY2024 driven by NPA resolutions, lower credit costs, and top-line growth.
- **Normalized Run-Rate (FY2025–FY2026):** Rose moderately to 40.0% by FY2026, though lower employee costs successfully offset this increase to keep net profit margins above 16%.

4.2. Contribution of Other Income to Sales



Non-interest revenue (Other Income)—comprising fee and commission income, forex and derivatives revenue, recovery from written-off accounts, and treasury gains (including

consolidated subsidiary contributions such as insurance and asset management where applicable)—acts as a critical counter-cyclical buffer and profitability enhancer for SBI.

Financial Year	Other Income / Sales (%)	Other Income / Total Income (%)	Core Sales / Total Income Share (%)	Revenue Mix Regime
FY2017	29.6%	22.8%	77.2%	Baseline pre-clean-up mix
FY2018	33.9%	25.3%	74.7%	Subsidiary stake monetization & fee buffer
FY2019	30.5%	23.4%	76.6%	Post-merger core interest recovery
FY2020	36.4%	26.7%	73.3%	Treasury windfalls & recovery surge
FY2021	38.6%	27.8%	72.2%	Low-rate environment treasury support
FY2022	40.3%	28.7%	71.3%	Peak non-interest revenue diversification
FY2023	34.9%	25.9%	74.1%	Core interest surge (+21% Sales) + bond yield rise
FY2024	35.4%	26.1%	73.9%	High-rate core expansion (+25.2% Sales)
FY2025	35.3%	26.1%	73.9%	Equilibrium between interest & fee growth
FY2026	38.4%	27.7%	72.3%	Fee, cross-sell & treasury re-acceleration

Mathematical Relationship Between the Two Metrics

- **The Income Identity:** Total Income is the sum of Sales (Interest Earned) and Other Income, mathematically linking the two ratios ($OtherIncome/TotalIncome = \frac{OtherIncome/Sales}{1+(OtherIncome/Sales)+}$).
- **Decade Average:** Other Income consistently contributed over a quarter of Total Income (~26.1%) and roughly 35.3% of Sales across FY2017–FY2026, buffering the bank against interest-rate volatility.

Counter-Cyclical Buffering (FY2017–FY2022)

- **Early Volatility (FY2017–FY2019):** Non-interest income cushioned operating profits during FY2018 when core Sales contracted by -0.6%, with the Other Income/Sales ratio peaking at 33.9%.
- **Pandemic & Rate Cuts (FY2020–FY2022):** As interest rates dropped and core Sales slowed, Other Income became the core growth engine—driven by treasury gains on bonds, fees, and recoveries—pushing the ratio to a 10-year peak of 40.3% in FY2022.

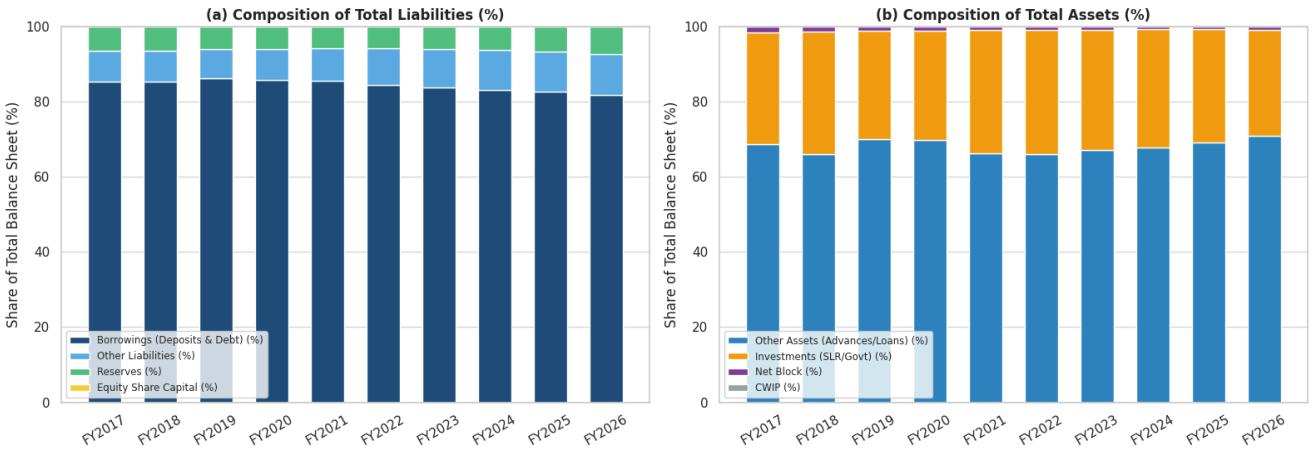
Normalization & Re-acceleration (FY2023–FY2026)

- **Rate Hiking Cycle (FY2023–FY2025):** As RBI hiked rates and core interest income surged, the Other Income/Sales ratio stabilized around 34.9%–35.4%, showing absolute Other Income grew in tandem with massive core lending expansion.
- **Late-Cycle Breakout (FY2026):** With core Sales growth moderating to 5.0%, Other Income rose to 38.4% of Sales, neutralizing operational expense spikes and helping SBI achieve a record Net Profit Margin of 16.2%.

5. BALANCE SHEET AND CASH FLOW

5.1. Changes in Asset and Liability Composition

Over the ten-year review period from FY2017 to FY2026, the State Bank of India (SBI) underwent a significant structural evolution in its balance sheet, marked by capital strengthening, aggressive deleveraging, and a strategic reallocation toward core lending activities following pandemic-era liquidity surpluses.



Structural Shifts in Total Liabilities and Capital Adequacy

- **Borrowings (Deposits & Debt):** Declined from ~85.5%–86.2% (FY2017–FY2019) to a decade-low of ~81.8% in FY2026 due to improved internal capital generation.
- **Other Liabilities & Provisions:** Expanded steadily from ~8.0% (FY2017–FY2019) to ~10.7% (FY2025–FY2026), reflecting stronger provisioning buffers and operational scale.
- **Equity & Reserves:** Equity share capital remained negligible (<0.2%), while reserves and surplus recovered from an NPA-merger dip (~5.6% in FY2021–FY2022) to ~7.3%–7.5% by FY2026, driven by high profitability.

Asset Deployment and Portfolio Rebalancing

- **Core Advances & Loans:** Recovered from pandemic lows (~66.0% in FY2022) to reach a decade-high of ~71.0% in FY2026, fueled by strong retail, agriculture, MSME, and corporate demand.
- **Investments (SLR & G-Secs):** Moved inversely to loans, peaking at ~32.8%–33.0% during the pandemic (FY2021–FY2022) before dropping to ~28.0% by FY2026 as SBI funded credit growth.
- **Fixed Assets:** Decreased steadily from ~1.5% (FY2017) to under 0.9% (FY2026), reflecting a strategic shift toward a capital-light digital banking architecture (e.g., YONO).

Financial Leverage Evolution and De-escalation

The structural fortification of SBI's balance sheet is best captured by the dramatic two-phase trajectory of its financial leverage multiples between FY2017 and FY2026:

Fiscal Year	Borrowings / Net Worth (x)	Total Assets / Net Worth (x)	Structural Phase & Strategic Drivers
FY2017	13.52x	~15.84x	Baseline pre-consolidation capital structure.
FY2018	13.42x	~15.70x	Associate bank merger absorption & initial NPA recognition.
FY2019	14.30x	~16.58x	Elevated provisioning eroding net worth growth; rising leverage.
FY2020	14.37x	~16.72x	Pre-pandemic balance sheet expansion outpacing internal accruals.
FY2021	15.06x	~17.58x	**Peak Leverage:** Surge in pandemic deposit inflows vs. moderate equity base.
FY2022	14.85x	~17.54x	Stabilization of asset quality; onset of profitability inflection.
FY2023	13.90x	~16.58x	Rapid internal capital generation via record net profits; active deleveraging.
FY2024	13.51x	~16.23x	Return to FY2017 leverage levels on a vastly larger balance sheet.
FY2025	12.42x	~15.01x	Sustained ROE expansion; organic net worth compounding outpaces debt growth.
FY2026	11.44x	~13.97x	**Decade-Low Leverage:** Exceptional solvency cushion and self-sustaining growth.

1. **The Leverage Escalation Phase (FY2017–FY2021):** Between FY2017 and FY2021, SBI's Borrowings-to-Net Worth multiple climbed from 13.52x to a peak of 15.06x, while the broader Total Assets-to-Net Worth equity multiplier rose from ~15.84x to ~17.58x. This upward drift was not driven by reckless borrowing, but rather by denominator compression: heavy credit provisioning for legacy corporate NPAs between FY2018 and FY2020 constrained net profit retention, while a massive influx of precautionary retail deposits during the FY2021 pandemic inflated total liabilities and assets faster than net worth could compound.

2. **The De-escalation and Solvency Fortification Phase (FY2021–FY2026):** Post-FY2021, SBI executed an extraordinary balance sheet de-risking. As credit costs normalized and Net Interest Margins (NIMs) expanded, surging net profits flowed directly into reserves. Consequently, net worth compounded at a rate significantly exceeding deposit and asset growth. The Borrowings-to-Net Worth ratio fell by 3.62 turns—dropping from 15.06x in FY2021 to 13.90x in FY2023, 12.42x in FY2025, and reaching 11.44x in FY2026. Simultaneously, Total Assets-to-Net Worth compressed from ~17.58x to ~13.97x. Equally notable is the widening spread between the two multiples (expanding from ~2.32x in FY2017 to ~2.53x in FY2026), confirming that

non-borrowing liabilities (float and provisions) and internal equity reserves are funding a progressively larger proportion of the bank's asset base.

5.2. Operating, Investing, and Financing Cash Flow Analysis

In commercial banking, cash flow statements behave differently than in non-financial corporates: customer deposits represent operating inflows, whereas loan disbursements and statutory securities purchases represent operating outflows. Consequently, negative operating cash flow in a banking context often signals vigorous credit expansion rather than operational distress, whereas large positive operating cash flows typically reflect deposit surges during periods of muted credit demand.

Granular Three-Way Cash Flow Breakdown

Fiscal Year	Operating Cash Flow (₹ Thousand Cr)	Investing Cash Flow (₹ Thousand Cr)	Financing Cash Flow (₹ Thousand Cr)	Net Cash Flow (₹ Thousand Cr)	Dominant Cash Flow Regime
FY2017	~76.0	~ -4.0	~ -4.0	~68.0	Demonetization-led deposit surge; high liquidity accumulation.
FY2018	-96.5	~12.5	~5.0	~ -77.5	**NPA Shock & Credit Contraction:** Post-demonetization deposit withdrawal & merger realignment.
FY2019	~29.0	~0.5	~0.5	~30.0	Liquidity stabilization and calibrated balance sheet repair.
FY2020	~24.0	~ -0.5	~5.0	~28.5	Steady pre-pandemic deposit accretion and subordinated debt raising.
FY2021	~89.5	~ -3.5	~7.0	~93.0	**Peak Liquidity Inflow:** Pandemic precautionary savings surge amidst muted loan demand.
FY2022	~57.0	~ -2.0	~ -3.5	~51.5	Continued strong deposit mobilization; early-stage credit recovery.
FY2023	-86.0	~ -1.0	~6.0	~ -80.0	**Massive Credit Expansion:** Aggressive deployment of accumulated cash into high-yield loans.
FY2024	~21.0	~ -3.5	~ -9.0	~8.5	Normalized operating cash generation; active debt retirement & dividend payouts.
FY2025	~45.0	~ -3.0	~ -11.0	~31.0	Strong deposit franchise scaling coupled with sustained capital distribution.

FY2026	~37.5	~9.0	~ -3.5	~44.0	Balanced core operating surplus augmented by strategic investment realizations.

Operating Cash Flow (OCF) Dynamics

- **FY2017–FY2018 (Volatility):** OCF spiked in FY17 due to demonetization deposits, but heavily reversed to –₹96,500 crore in FY18 due to remonetization withdrawals, bank mergers, and heavy NPA provisioning.
- **FY2019–FY22 (Pandemic Liquidity):** OCF grew steadily, peaking at ₹89,500 crore in FY21 as pandemic-driven savings surged and loan demand stalled.
- **FY2023 (Credit Super-Cycle):** OCF dropped to –₹86,000 crore as SBI intentionally deployed its excess pandemic liquidity into higher-yielding retail and corporate loans to capture market share.
- **FY2024–FY2026 (Mature Equilibrium):** OCF stabilized at positive levels (up to ₹45,000 crore), with robust profits and balanced deposit mobilization supporting credit growth.

Investing Cash Flow (ICF) Dynamics

- **Routine Capex (FY17, FY20–FY25):** Modest, consistent outflows (–₹500 to –₹4,000 crore annually) driven by disciplined spending on IT and branch modernization.
- **Strategic Realizations (FY18, FY26):** Major positive inflows (₹12,500 crore in FY18; ₹9,000 crore in FY26) from non-core asset sales, treasury gains, and subsidiary value unlocking.

Financing Cash Flow (FCF) Dynamics

- **Capital Raising (FY18–FY21, FY23):** Positive inflows driven by AT1/Tier-2 bond issuances, equity raises (QIP), and suspended dividends to build capital buffers.
- **Deleveraging & Payouts (FY24–FY26):** Cash flows turned negative as strong internal profits allowed SBI to redeem high-cost legacy bonds, pay down debt, and resume high shareholder dividends.

Balance Sheet and Cash Flow Interlinkages

- **FY2017–FY2021:** Asset-quality issues and pandemic deposits led to high leverage, with excess liquidity parked in lower-yielding sovereign bonds (33% of assets).
- **FY2022–FY2026:** SBI successfully shifted excess liquidity into higher-yielding customer loans (reaching 71% of assets by FY26). Strong profitability reduced financial leverage to a decade-low (11.44x), funding self-sustaining growth, debt reduction, and consistent net cash surpluses.

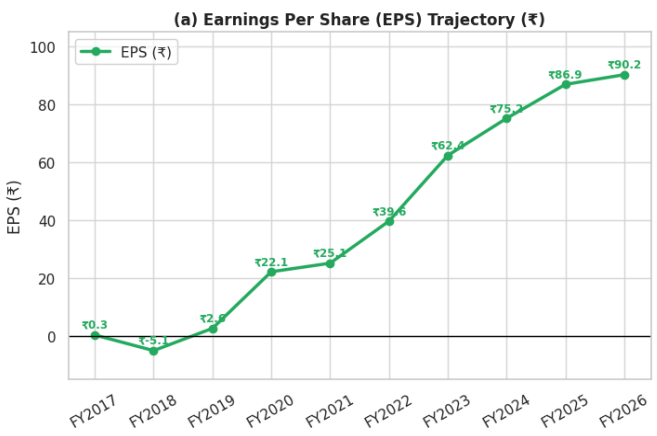
6. SHAREHOLDER RETURNS

Over the ten-year period from FY2017 to FY2026, the shareholder return profile of the State Bank of India (SBI) underwent a structural transformation. Moving from a phase of severe balance sheet stress and earnings compression during the late-2010s asset quality review and associate bank merger cycle, the bank transitioned into an era of sustained profitability, consistent book value expansion, regular dividend distribution, and substantial capital appreciation.

Financial Year	EPS (₹) PNG	Normalized P/E (x) PNG	P/B Ratio (x) PNG	DPS (₹) PNG	Implied Payout Ratio (%)	Market Share Price (₹) PNG
FY2017	0.3	—	1.08x	2.6	—	293
FY2018	-5.1	—	0.97x	0.0	0.0%	250
FY2019	2.6	—	1.22x	0.0	0.0%	321
FY2020	22.1	8.9x	0.70x	0.0	0.0%	197
FY2021	25.1	14.5x	1.18x	4.0	15.9%	364
FY2022	39.6	12.5x	1.44x	7.1	17.9%	494
FY2023	62.4	8.4x	1.30x	11.3	18.1%	524
FY2024	75.2	10.0x	1.62x	13.7	18.2%	752
FY2025	86.9	8.9x	1.41x	15.9	18.3%	772
FY2026	90.2	10.9x	1.52x	17.3	19.2%	979

6.1 Trends in Earnings Per Share (EPS), Price-to-Earnings (P/E), and Price-to-Book (P/B) Ratios

Earnings Per Share (EPS) Trajectory

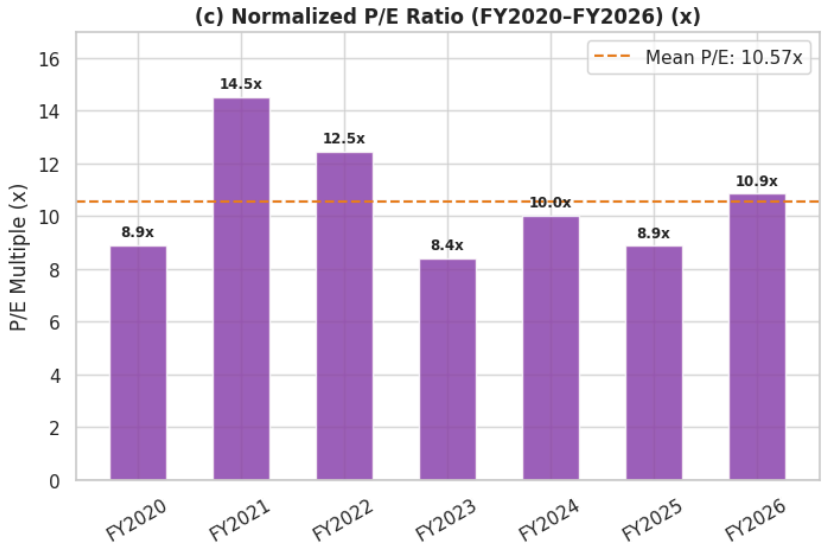


The ten-year earnings trajectory of SBI can be segmented into three distinct operational phases:

1. **Asset Quality Cleanup and Merger Absorption (FY2017–FY2019):** EPS declined from ₹0.3 in FY2017 to −₹5.1 in FY2018 due to losses and NPA provisioning, before recovering to ₹2.6 in FY2019.

2. **Recovery and Pandemic Resilience (FY2020–FY2021):** EPS surged to ₹22.1 in FY2020 and grew by 13.6% to ₹25.1 in FY2021, reflecting improved profitability and resilience during the pandemic.
3. **Structural Expansion and Secular Compounding (FY2022–FY2026):** EPS rose from ₹39.6 in FY2022 to a record ₹90.2 in FY2026, driven by credit growth and improved profitability. From FY2020 to FY2026, EPS grew more than fourfold, achieving an approximate CAGR of 26.4%.

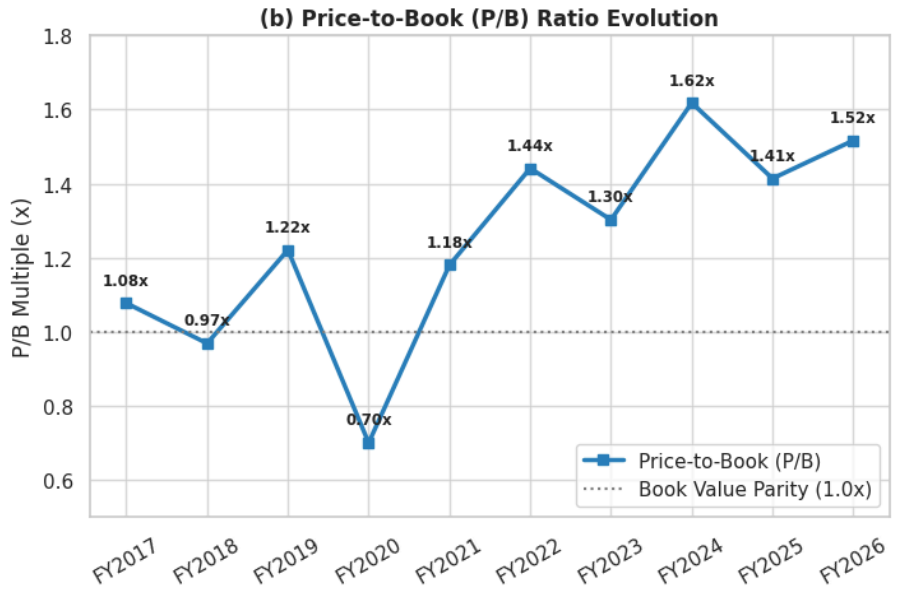
Normalized Price-to-Earnings (P/E) Ratio Analysis



Because near-zero and negative earnings between FY2017 and FY2019 distort traditional earnings multiples, the normalized P/E evaluation covers the profitable expansion cycle from FY2020 to FY2026, during which the bank traded at a mean P/E multiple of **10.57x**:

- **Multiple Expansion on Recovery Expectations (FY2020–FY2022):** In FY2020, despite strong earnings recovery, pandemic-induced market sell-offs compressed SBI's P/E to 8.9x. As clarity emerged regarding asset quality resilience in FY2021, equity markets re-rated the stock Ahead of earnings growth, pushing the P/E multiple to a peak of 14.5x in FY2021 and 12.5x in FY2022.
- **Earnings-Led Valuation Normalization (FY2023–FY2026):** Between FY2023 and FY2025, rapid bottom-line expansion outpaced share price appreciation, causing the P/E ratio to compress to 8.4x in FY2023, recover to 10.0x in FY2024, and settle at 8.9x in FY2025. In FY2026, sustained institutional confidence and stable return ratios supported a re-rating back above the historical average to **10.9x**. This stability around the 10.57x mean indicates that SBI's long-term stock rally has been fundamentally backed by earnings growth rather than speculative multiple expansion.

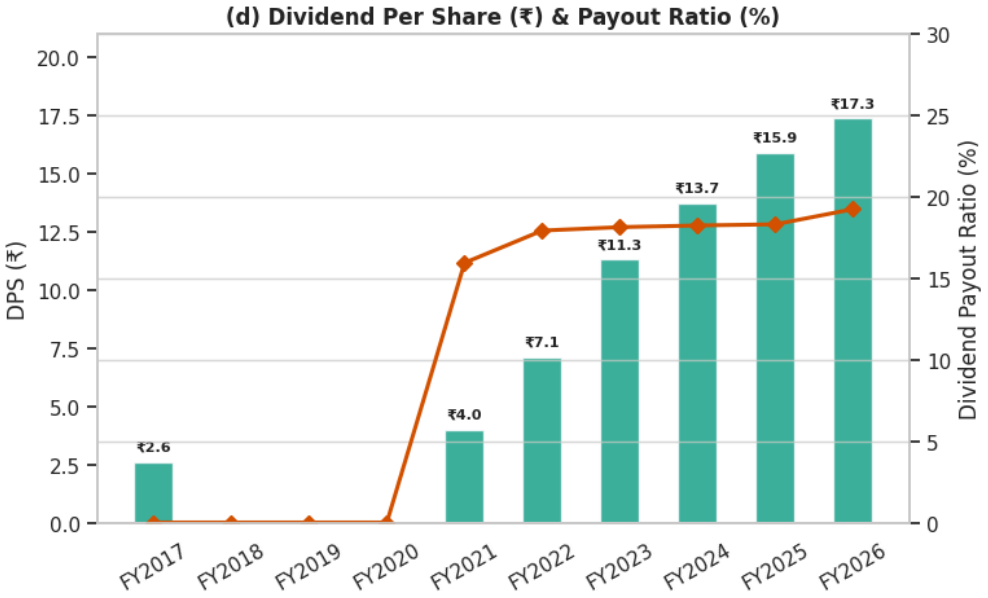
Price-to-Book (P/B) Ratio Evolution



For banking institutions, the Price-to-Book (P/B) ratio serves as a core indicator of market confidence in net worth quality and Return on Equity (RoE) sustainability relative to the cost of capital:

- **Oscillation Around Book Value Parity (FY2017–FY2020):** SBI's P/B ratio fluctuated from 1.08x in FY2017 to 0.97x in FY2018, recovered to 1.22x in FY2019, and fell to a period low of 0.70x in FY2020 amid pandemic-related market uncertainty.
- **Sustained Premium Valuation (FY2021–FY2026):** The P/B ratio recovered to 1.18x in FY2021 and peaked at 1.62x in FY2024, remaining elevated at 1.52x in FY2026. Trading consistently above book value indicates improved investor confidence in SBI's profitability and growth prospects.

6.2 Trends in Dividends and Dividend Payout Ratio



SBI's dividend policy over the FY2017–FY2026 decade reflects a disciplined balance between regulatory capital preservation and shareholder wealth distribution:

- **Capital Conservation and Dividend Suspension (FY2017–FY2020):** SBI's DPS declined from ₹2.6 in FY2017 to ₹0 in FY2018–FY2020 due to losses, capital conservation needs, and RBI's COVID-19 dividend restrictions.
- **Resumption and Dividend Growth (FY2021–FY2026):** Dividends resumed at ₹4.0 per share in FY2021, rising consistently to ₹17.3 in FY2026, representing a 34.0% CAGR over the period.
- **Stability of Dividend Payout Ratio:** The payout ratio remained conservative, increasing from 15.9% in FY2021 to 19.2% in FY2026. Retaining over 80% of profits supported loan growth while ensuring steadily rising shareholder returns.

6.3 Relationship Between Share Price and EPS

To evaluate the efficiency with which SBI's fundamental earnings growth translated into market capitalization gains over FY2017–FY2026, an ordinary least squares (OLS) linear regression analysis was conducted with Earnings Per Share (EPS) as the independent variable and Market Share Price as the dependent variable.

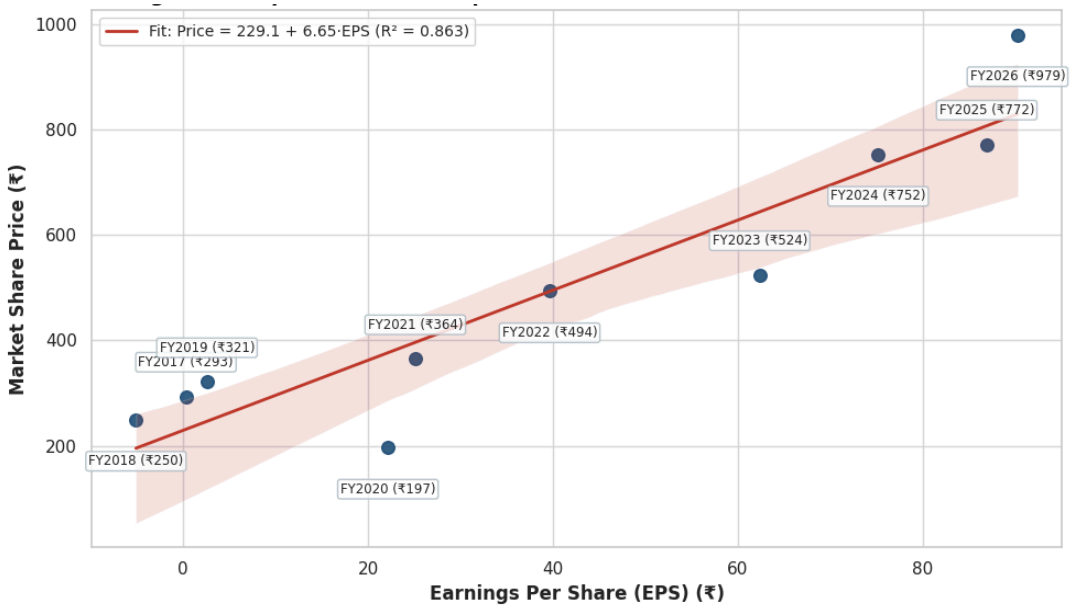
Empirical Regression Model

The empirical relationship over the ten-year period is governed by the following linear equation:

$$MarketSharePrice(₹) = 229.1 + 6.65 \times EPS(₹)$$

- **Coefficient of Determination ($R^2 = 0.863$):** An R^2 value of **0.863** indicates that **86.3%** of the variance in SBI's market share price over the FY2017–FY2026 decade is directly explained by changes in its Earnings Per Share. The remaining 13.7% of price variation is attributable to exogenous macroeconomic factors, interest rate cycles, and broader market liquidity shifts.
- **Intercept ($\alpha = ₹229.1$):** The positive intercept of ₹229.1 represents the baseline intrinsic floor value assigned by the market to SBI's sovereign-backed franchise, extensive deposit base, and subsidiary ecosystem even during periods of zero standalone net profit. This is empirically corroborated by FY2017–FY2019 trading levels, where the stock remained resilient between ₹250 and ₹321 despite EPS hovering between ₹-5.1 and ₹2.6.
- **Slope Coefficient ($\beta = 6.65$):** The slope coefficient demonstrates that, on average across the full ten-year transition cycle, every **₹1.00 increase in EPS generated an incremental ₹6.65 in market share price**.

Historical Price-to-Earnings Alignment and Residual Analysis



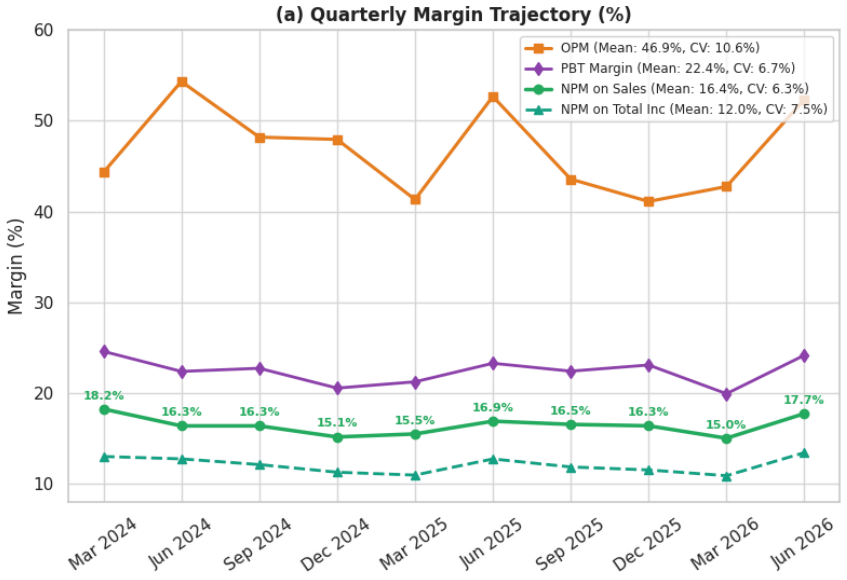
1. **FY2017–FY2019 (Downside Cushion):** Despite a sharp EPS drop to negative ₹5.1 in FY18, the share price only declined 14.7% (to ₹250). Prices remained above the regression trendline as investors looked past legacy provisioning to focus on long-term franchise value.
2. **FY2020 (Pandemic Anomaly):** Marked the largest negative outlier below the trendline. Although EPS jumped to ₹22.1 (predicting a ₹376 share price), COVID-19 risk aversion dragged the actual price down to a decade-low of ₹197.
3. **FY2021–FY2025 (Mean Reversion):** Share prices rapidly realigned with earnings fundamentals, closely tracking predicted models by hitting ₹494 in FY22. A valuation lag in FY23 (₹524 actual vs. ₹644 predicted) triggered a 43.5% catch-up surge in FY24 (to ₹752), before consolidating at ₹772 in FY25.
4. **FY2026 (Breakout & Re-Rating):** The stock broke above the trendline, hitting ₹979 against a predicted ₹829. This positive divergence reflects a structural market re-rating driven by consistent earnings and dividend growth, capping a 234.1% decade-long appreciation and a nearly fivefold recovery from the FY20 trough.

7. QUARTERLY ANALYSIS

This chapter presents a systematic review of the quarterly financial performance of the State Bank of India (SBI) across ten consecutive quarters from March 2024 to June 2026. The analysis is structured into two primary dimensions: an evaluation of absolute progression and quarter-on-quarter (QoQ) growth volatility across sales, operating profit, and net profit, followed by an assessment of profit margin trajectories, statistical dispersion, and bottom-line stability.

7.1 Trends in Quarterly Sales, Operating Profit, and Net Profit

Quarterly Progression in Revenue and Profitability

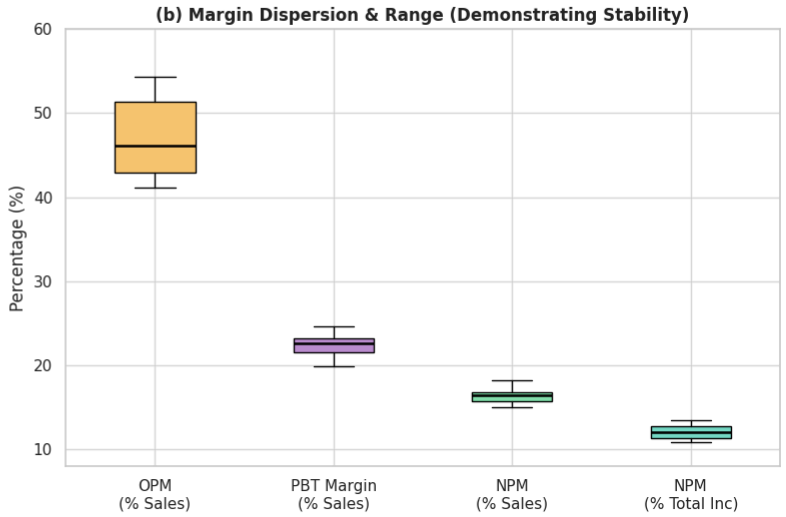


Between March 2024 and June 2026, State Bank of India demonstrated a consistent structural expansion in its top-line revenue alongside cyclical movements in operating profitability and a resilient net profit trajectory.

Quarter	Sales (₹ Thousand Cr)	Operating Profit (₹ Thousand Cr, Approx. Trajectory)	Net Profit (₹ Thousand Cr)
Mar 2024	117.5	~52.2	21.4
Jun 2024	118.2	~64.2	19.3
Sep 2024	121.0	~58.3	19.8
Dec 2024	124.7	~59.7	18.9
Mar 2025	126.8	~52.4	19.6
Jun 2025	125.7	~66.2	21.2
Sep 2025	128.0	~55.7	21.1
Dec 2025	130.4	~53.6	21.3
Mar 2026	131.1	~56.1	19.6
Jun 2026	136.2	~71.4	24.1

- **Secular Expansion in Sales:** Quarterly sales increased from ₹117.5 thousand crores in March 2024 to ₹136.2 thousand crores in June 2026, a growth of 15.9%, with only a minor temporary decline in June 2025.
- **Seasonal Cyclicality in Operating Profit:** Operating profit fluctuated between approximately ₹52 thousand crores and ₹71 thousand crores, consistently surging in June quarters and moderating during subsequent quarters, reaching a period high in June 2026.
- **Bottom-Line Consolidation and Breakout in Net Profit:** Net profit remained between ₹18.9 thousand crores and ₹21.4 thousand crores during the first nine quarters, before recovering from ₹19.6 thousand crores in March 2026 to a period high of ₹24.1 thousand crores in June 2026.

Quarter-on-Quarter (QoQ) Growth Volatility



Comparing the sequential percentage changes across sales, operating profit, and net profit reveals a pronounced divergence between top-line steadiness and mid-line operational swings:

- **Low Volatility in Sales Growth:** QoQ sales growth remained relatively stable, rising from below +1% in June 2024 to +3% in December 2024, briefly falling to −0.9% in June 2025, and reaching a period high of +3.9% in June 2026.
- **High Amplitude in Operating Profit Growth:** Operating profit growth showed sharp cyclical fluctuations, ranging from −15.8% to +27.2%. It consistently peaked in June quarters before experiencing significant sequential declines.
- **Buffered Volatility in Net Profit Growth:** Net profit growth was comparatively stable, fluctuating between −9.8% and +8.2% during much of the period, with near-flat growth in September and December 2025. After falling 8.0% in March 2026, it surged by 23.0% in June 2026.

7.2 Quarterly Profit Margin Stability

Quarterly Margin Trajectory

To evaluate how effectively top-line revenue translated into earnings across different stages of the income statement, four profitability ratios were tracked from March 2024 to June 2026: Operating Profit Margin (OPM), Profit Before Tax (PBT) Margin, Net Profit Margin (NPM) on Sales, and Net Profit Margin (NPM) on Total Income.

Profitability Metric	Mean (%)	Coefficient of Variation (CV %)	Observed Trajectory & Range Characteristics
Operating Profit Margin (OPM % Sales)	46.9%	10.6%	Cyclical peaks above 52%–54% in June quarters; troughs near 41% in Mar 2025 and Dec 2025.
PBT Margin (% Sales)	22.4%	6.7%	Stable corridor between ~19.8% (Mar 2026) and ~24.6% (Mar 2024).
NPM on Sales (% Sales)	16.4%	6.3%	Highest relative stability; ranges strictly between 15.0% (Mar 2026) and 18.2% (Mar 2024).
NPM on Total Income (% Total Inc)	12.0%	7.5%	Parallel trajectory to NPM on Sales, bounded tightly between ~10.8% and ~13.5%.

- **Operating Profit Margin (OPM) Dynamics:** OPM averaged 46.9% with a CV of 10.6%, showing the highest volatility. It peaked at 54.3% in June 2024, fell to 41.1% in December 2025, and recovered to 52.4% in June 2026.
- **Profit Before Tax (PBT) Margin Absorption:** PBT margin averaged 22.4% with a CV of 6.7%, indicating lower volatility than OPM. It declined from 24.6% in March 2024 to 19.8% in March 2026 before rebounding to 24.1% in June 2026.
- **Net Profit Margin (NPM) on Sales Precision:** NPM on Sales averaged 16.4%, with the lowest CV of 6.3%, indicating the greatest stability. It declined from 18.2% in March 2024 to 15.0% in March 2026 before recovering to 17.7% in June 2026.
- **Net Profit Margin (NPM) on Total Income Consistency:** NPM on Total Income averaged 12.0% with a CV of 7.5%. It fell from 13.0% in March 2024 to 10.8% in March 2026, then rose to a period high of 13.5% in June 2026.

Margin Dispersion and Structural Stability

- Wide Operational Spread: OPM shows the widest dispersion, with a median of 46.1% and a range of 41.1%–54.3%, reflecting quarterly cost and provisioning fluctuations.
- Progressive Variance Compression: PBT margin has a narrower distribution, centered around 22.5%, with a total range of less than 5 percentage points.
- Stable Bottom-Line Profitability: NPM on Sales clusters around 16.3%, within a 15.0%–18.2% range, while NPM on Total Income remains near 12.0%, within an approximate 2.7-percentage-point range. This indicates strong net profit stability and earnings predictability despite operating fluctuations during March 2024–June 2026.

8. CONCLUSION, LIMITATIONS AND FUTURE SCOPE

SBI's financial performance from FY2017–FY2026 shows a transformation from financial stress and losses to revenue growth, improved efficiency, lower leverage, higher profitability, and shareholder value creation.

Key Financial Summary

Indicator	FY2017	Trough/Peak	FY2026
Annual Sales / Interest Earned	₹230.4k Cr	₹229.0k Cr (FY2018)	₹514.9k Cr
Interest Expense / Sales	64.7%	53.9% (FY2022)	61.2%
Employee Cost / Sales	15.5%	21.2% (FY2022)	14.1%
Other Income / Total Income	22.8%	28.7% (FY2022)	27.7%
Borrowings / Net Worth	13.52x	15.06x (FY2021)	11.44x
EPS	₹0.3	-₹5.1 (FY2018)	₹90.2
Share Price / P/B	₹293 / 1.08x	₹197 / 0.70x (FY2020)	₹979 / 1.52x
Dividend Per Share	₹2.6	₹0 (FY2018–FY2020)	₹17.3
Quarterly Net Profit	₹21.4k Cr (Mar 2024)	₹18.9k Cr (Dec 2024)	₹24.1k Cr (Jun 2026)

Other key findings: revenue increased 123.5% over the decade; employee-cost efficiency improved by 710 basis points from FY2022 to FY2026; and quarterly net profit margin remained relatively stable, averaging 16.4% with a 6.3% coefficient of variation (CV).

8.1. Conclusions

A. Two-Phase Revenue Expansion and Operating Leverage

- Revenue Growth: Sales grew from ₹230.4k Cr in FY2017 to ₹290.0k Cr in FY2022 (+25.9%), then increased 77.6% to ₹514.9k Cr in FY2026.
- Funding Costs: Interest expense/sales declined from 64.7% in FY2017 to 53.9% in FY2022, before rising to 61.2% in FY2026 due to deposit repricing.
- Operating Efficiency: Other operating expenses rose sharply in FY2018 due to legacy NPA provisioning, then declined. Employee cost/sales fell from 21.2% in FY2022 to 14.1% in FY2026, showing improved operating leverage.
- Other Income: Its share of total income increased from 22.8% in FY2017 to 28.7% in FY2022 and remained strong at 27.7% in FY2026.

B. Balance Sheet Strengthening, De-leveraging and Cash Flow Cycles

- Lower Leverage: Borrowings/net worth declined from a peak of 15.06x in FY2021 to 11.44x in FY2026; total assets/net worth was approximately 17.6x in FY2021.
- Capital Accumulation: Borrowings, including deposits and debt, funded approximately 82–86% of assets. Reserves and other liabilities gained share, while advances recovered

from approximately 66% of assets in FY2021–FY2022 to 71% in FY2026. Government investments had reached around 32% during FY2021–FY2022.

- Cash Flow Cycles: Operating cash flow was -₹96.5k Cr in FY2018 due to credit contraction and NPAs, and -₹86.0k Cr in FY2023 due to rapid credit expansion. Operating and net cash flows remained positive from FY2024–FY2026.

C. Quarterly Margin Stability vs. Operating Profit Seasonality (Mar 2024–Jun 2026)

- Revenue and Profit: Across 10 quarters, sales increased from ₹117.5k Cr to ₹136.2k Cr, with a minor dip to ₹125.7k Cr in June 2025. Net profit remained between ₹18.9k–₹21.4k Cr for nine quarters before reaching ₹24.1k Cr in June 2026.
- Margin Stability: Operating profit quarterly growth fluctuated from approximately -16% to +27%. Average OPM was 46.9% (CV 10.6%), PBT margin 22.4% (CV 6.7%), NPM on sales 16.4% (CV 6.3%, range 15.0–18.2%), and NPM on total income 12.0% (CV 7.5%).

D. Shareholder Value Creation and Market Re-rating

- Earnings Recovery: EPS recovered from -₹5.1 in FY2018 to ₹90.2 in FY2026, rising from ₹22.1 in FY2020.
- Share Price Relationship: OLS regression found a strong EPS–share-price relationship:

$$Price = 229.1 + 6.65(EPS), R^2 = 0.863$$

EPS explains 86.3% of share-price variation in the sample.

- Valuation: P/B recovered from 0.70x in FY2020 to above 1.0x from FY2021, peaking at 1.62x in FY2024 and reaching 1.52x in FY2026. Normalized P/E averaged 10.57x during FY2020–FY2026, ranging from 8.4x to 14.5x and ending at 10.9x. This suggests share-price growth was mainly earnings-driven.
- Dividends: After suspension during FY2018–FY2020, dividends resumed at ₹4 per share in FY2021 and increased to ₹17.3 in FY2026, with a payout ratio of approximately 18–19% since FY2022.

8.2. Limitations

A. Non-Standard Banking Accounting Terminology and Aggregation

- Generic Accounting Labels: “Sales” represents interest earned, “Borrowings” combines deposits and wholesale debt, and “Other Assets” represents advances/loans. Combining CASA, term deposits and Tier-1/Tier-2 bonds hides funding costs and liability stability.
- Missing Banking Metrics: NII, NIM, cost of deposits, yield on advances, GNPA, NNPA, slippage ratio and PCR are unavailable. Other operating expenses may combine

overheads, provisions and contingencies, making operational efficiency difficult to isolate.
- Unorthodox OPM Definition: OPM averages 46.9%, versus PBT margin of 22.4%. Including other income in the numerator but excluding it from sales, or using a pre-interest/pre-provision convention, may overstate operating margins.

B. Statistical and Methodological Constraints

- Small Regression Sample: The EPS–price model uses only 10 annual observations (FY2017–FY2026). Non-stationarity and positive serial correlation may inflate R^2 and narrow confidence intervals. FY2020 (₹197) and FY2026 (₹979) are notable outliers.
- P/E Selection Bias: The FY2020–FY2026 normalized P/E average of 10.57x excludes FY2017–FY2019 because near-zero or negative EPS produces extreme or negative P/E ratios, omitting the distressed valuation period.
- Limited Quarterly Period: Only 10 quarters (Mar 2024–Jun 2026) are analysed. Low NPM variation (CV 6.3%) reflects a single relatively benign credit cycle, not a complete interest-rate or default cycle.
- Cash Flow Distortion: Loan disbursements and deposit inflows are classified as operating cash flows. Thus, negative cash flow may indicate either distress (FY2018: -₹96.5k Cr) or strong credit expansion (FY2023: -₹86.0k Cr).

8.3. Future Scope

A. Granular Banking-Specific Decomposition

- Funding Spread Analysis: Separate CASA, retail term-deposit and bulk-borrowing costs; compare them with yields on advances and investments to assess NIM sensitivity to interest-rate cycles.
- Expense and Credit-Cost Separation: Break down other operating expenses and the OPM–PBT margin gap into loan-loss provisions, treasury gains/losses and core administrative expenses.
- Capital Adequacy: Add Basel III CET-1, Tier-1 and total CRAR ratios, along with risk-weighted asset (RWA) density, to assess growth capacity before equity dilution is needed.

B. Econometric and Valuation Model Enhancements

- Advanced Valuation Models: Expand EPS–share-price regression using ROE, P/B drivers, GNPA ratios and 10-year Government Security yields. Use quarterly observations to overcome the small-sample limitation.
- Seasonality Analysis: Investigate recurring June-quarter operating-profit growth spikes in 2024, 2025 and 2026 using seasonal-trend decomposition, such as X-13ARIMA-SEATS, to identify treasury/fee recognition and Q4 provisioning base effects.

C. Peer Benchmarking and Stress Testing

- Bank Comparison: Compare SBI's leverage reduction (15.06x to 11.44x), employee-cost ratio (14.1%) and P/E–EPS growth with leading private and public sector banks to assess relative valuation premiums or discounts.
- Macroeconomic Stress Testing: Model the FY2026 balance sheet (advances approximately 71% of assets; interest expense/sales 61.2%) under:
    - RBI policy-rate cuts of 50–100 basis points.
    - A 50-basis-point increase in credit slippages.